RTX 3090 Local Inference Cost Analysis

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Executive Summary

This document provides a detailed, empirically-verified cost analysis for running local LLM inference on an NVIDIA RTX 3090 GPU via llama.cpp, compared to cloud API pricing. All figures are based on **measured benchmarks** from the production server, not estimates.

Bottom line: Local inference costs **$0.04/M input tokens** and **$0.15/M output tokens** — a **99.7–99.8% savings** over Claude Opus 4.6 cloud pricing.

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1. Hardware Configuration

Component	Detail
GPU	NVIDIA RTX 3090 (24GB VRAM)
Acquisition Cost	~$900 (purchased April 2025)
TDP	350W (spec), ~300W sustained inference load
Host	`olla` (VM 101 on Proxmox)
Software	llama.cpp (llama-server)
Model	Qwen 3.5 (GGUF quantized)
Context Window	262,144 tokens
Location	Louisiana, USA

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2. Measured Performance

Benchmarks were taken directly from the llama-server `/v1/chat/completions` endpoint with timing data on April 5, 2026.

Prefill (Input/Prompt Processing)

- **Speed:** 493.7 tokens/second
- **Workload type:** Compute-bound, parallel batch processing
- **What it does:** Processes the entire input prompt through the model in one pass

Decode (Output/Token Generation)

- **Speed:** 122.5 tokens/second
- **Workload type:** Memory-bandwidth-bound, sequential (one token at a time)
- **What it does:** Generates response tokens autoregressively

Why They Differ

Prefill processes all input tokens simultaneously using the GPU's parallel compute cores (tensor cores, CUDA cores). Decode generates tokens one at a time, bottlenecked by memory bandwidth as it reads the full KV cache for each token. This is why prefill is ~4x faster than decode on the 3090.

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3. Electricity Cost

Parameter	Value
Electricity Rate	$0.155/kWh (Louisiana residential, verified)
GPU Power Draw (inference)	~300W sustained (measured average under load)
GPU Power Draw (idle)	~15-25W

Cost Per Million Input Tokens (Prefill)

```
Time = 1,000,000 tokens ÷ 493.7 tok/s = 2,025.5 seconds = 0.5627 hours
Energy = 0.300 kW × 0.5627 hours = 0.1688 kWh
Cost = 0.1688 kWh × $0.155/kWh = $0.0262
```

Electricity cost per 1M input tokens: $0.026

Cost Per Million Output Tokens (Decode)

```
Time = 1,000,000 tokens ÷ 122.5 tok/s = 8,163.3 seconds = 2.2676 hours
Energy = 0.300 kW × 2.2676 hours = 0.6803 kWh
Cost = 0.6803 kWh × $0.155/kWh = $0.1054
```

Electricity cost per 1M output tokens: $0.105

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4. Hardware Depreciation

Using straight-line depreciation over 5 years (20% annual rate):

```
Annual depreciation = $900 × 20% = $180/year
Hourly depreciation = $180 ÷ 8,760 hours/year = $0.02055/hour
```

Depreciation Per Million Input Tokens

```
Time = 0.5627 hours
Cost = 0.5627 × $0.02055 = $0.0116
```

Depreciation per 1M input tokens: $0.012

Depreciation Per Million Output Tokens

```
Time = 2.2676 hours
Cost = 2.2676 × $0.02055 = $0.0466
```

Depreciation per 1M output tokens: $0.047

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5. Total Cost Per Million Tokens

Cost Component	Input (per 1M)	Output (per 1M)
Electricity	$0.026	$0.105
Hardware Depreciation	$0.012	$0.047
Total	**$0.038**	**$0.152**
Rounded (for config)	**$0.04**	**$0.15**

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6. Cloud Comparison

vs. Claude Opus 4.6 (Anthropic)

Metric	Local (3090)	Cloud	Savings
Input per 1M tokens	$0.04	$15.00	99.73%
Output per 1M tokens	$0.15	$75.00	99.80%
Blended (50/50 I/O)	$0.095	$45.00	99.79%

vs. GPT-5-mini (OpenAI)

Metric	Local (3090)	Cloud	Savings
Input per 1M tokens	$0.04	$0.30	86.7%
Output per 1M tokens	$0.15	$1.25	88.0%

vs. GPT-5.2 Codex (OpenAI)

Metric	Local (3090)	Cloud	Savings
Input per 1M tokens	$0.04	$2.50	98.4%
Output per 1M tokens	$0.15	$10.00	98.5%

Monthly Projection (estimated 75M tokens/month, 60% input / 40% output)

Provider	Input Cost	Output Cost	Total Monthly
Local (3090)	$1.80	$4.56	**$6.36**

Provider	Input Cost	Output Cost	Total Monthly
Claude Opus 4.6	$675.00	$2,250.00	$2,925.00
GPT-5.2 Codex	$112.50	$300.00	$412.50

Annual savings vs Claude Opus 4.6: ~$35,024 Annual savings vs GPT-5.2 Codex: ~$4,874

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7. OpenClaw Configuration

The cost values are configured in `openclaw.json` under the model definition. These values are in **dollars per million tokens**:

```
{
  "id": "llamacpp",
  "name": "llamacpp (local)",
  "cost": {
    "input": 0.04,
    "output": 0.15,
    "cacheRead": 0,
    "cacheWrite": 0
  }
}
```

Path: `models.providers.olla.models[0].cost`

What These Values Do

- OpenClaw uses these to track and report usage costs in `/status` and cost reports
- The daily cost report cron job references these rates
- Monthly invoices are generated using these figures
- `cacheRead` and `cacheWrite` are $0 because KV cache operations are local with no additional cost

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8. Assumptions & Caveats

1. **Power draw is estimated at 300W** — actual draw varies by batch size, context length, and quantization. True measurement requires a power meter (Kill-A-Watt or similar).
2. **Depreciation is simplified** — 20% straight-line over 5 years. GPU resale value may differ. A $900 GPU in 2025 might retain $400-500 value after 2 years, or could be worth less if newer generations arrive.
3. **Idle power not included** — If the server runs 24/7 regardless of inference load, idle power (~20W) is a fixed cost not attributable to per-token pricing. At $0.155/kWh, idle costs ~$27/year.

4. **No cooling/overhead** — Doesn't include AC costs for heat dissipation, network power, or host system power draw (CPU, RAM, PSU efficiency loss).
5. **Model-dependent** — These numbers are specific to Qwen 3.5 GGUF on this hardware. Larger models (70B+) would be slower and cost more per token. Smaller models would be faster and cheaper.
6. **Cloud comparison uses list pricing** — Volume discounts, committed use, or batch API pricing would narrow the gap somewhat.

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9. Metrics & Automated Tracking

As of April 5, 2026, the full metrics pipeline is operational:

Prometheus Metrics Endpoint

- **URL:** `http://olla:11434/metrics`
- **Enabled:** `--metrics` flag added to llama-server.service
- **Available metrics:**

Metric	Type	Description
`llamacpp:prompt_tokens_total`	counter	Cumulative input tokens processed
`llamacpp:tokens_predicted_total`	counter	Cumulative output tokens generated
`llamacpp:prompt_seconds_total`	counter	Total prefill compute time
`llamacpp:tokens_predicted_seconds_total`	counter	Total decode compute time
`llamacpp:prompt_tokens_seconds`	gauge	Current avg prefill throughput (tok/s)
`llamacpp:predicted_tokens_seconds`	gauge	Current avg decode throughput (tok/s)
`llamacpp:requests_processing`	gauge	Active requests
`llamacpp:requests_deferred`	gauge	Queued requests

Metrics Scraper

- **Script:** `scripts/olla-metrics-scraper.sh`
- **Usage:** `./olla-metrics-scraper.sh` (report) or `./olla-metrics-scraper.sh --reset` (new baseline)
- **Output:** JSON logs to `cost-tracking/daily/YYYY-MM-DD.json`
- **Cost formula:**
  - Input cost = `prompt_tokens_total` delta × $0.04 / 1,000,000
  - Output cost = `tokens_predicted_total` delta × $0.15 / 1,000,000

Automated Reporting (Cron Jobs)

Job	Schedule	What it does
Daily Olla Cost Report	12:00 UTC daily	Scrapes metrics, reports costs, resets baseline

Job	Schedule	What it does
Monthly Invoice	09:00 UTC, 1st of month	Aggregates daily data, generates PDF, emails to Jeremy

Data Flow

```
llama-server --metrics → olla:11434/metrics
    → olla-metrics-scraper.sh (daily cron)
        → cost-tracking/daily/YYYY-MM-DD.json
            → Daily report (Matrix announce)
            → Monthly invoice (PDF → email)
```

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10. Recommendations

1. **Install a Kill-A-Watt meter** on the GPU server to measure actual wall power during inference — would make the electricity calculation exact.
2. ~~**Enable --metrics flag**~~ **DONE** — Metrics endpoint live at olla:11434/metrics
3. **Consider idle cost** — If the server runs 24/7 regardless of inference load, idle power (~20W) is a fixed cost not attributable to per-token pricing. At $0.155/kWh, idle costs ~$27/year.
4. **Review quarterly** — Electricity rates, hardware value, and model performance change. Recalculate every 3 months.
5. **Connect to Prometheus/Grafana** (future) — The metrics endpoint is Prometheus-compatible. Could add dashboards for real-time throughput and cost visualization.

Appendix: Benchmark Raw Data

Test Date: April 5, 2026, 23:30 UTC **Server:** olla:11434 (llama-server) **Model:** qwen3.5:latest.gguf

Run 1 (Cold)

```
Prompt tokens: 28 | Completion tokens: 12
Prefill: 160.8 tok/s (174ms)
Decode: 107.6 tok/s (112ms)
```

Run 2 (Warm)

```
Prompt tokens: 30 | Completion tokens: 271
Prefill: 493.7 tok/s (61ms)
Decode: 122.5 tok/s (2,211ms)
```

Used warm-run values for cost calculations (representative of typical workload with warm KV cache).

Appendix B: OpenClaw Config (Applied)

The following cost values are live in `openclaw.json` as of April 5, 2026:

```
{
  "models": {
    "providers": {
      "olla": {
        "models": [{
          "id": "llamacpp",
          "cost": {
            "input": 0.04,
            "output": 0.15,
            "cacheRead": 0,
            "cacheWrite": 0
          }
        }]
      }
    }
  }
}
```

Appendix C: llama-server.service (Current)

```
ExecStart=/opt/llama.cpp/build/bin/llama-server \
  -m /opt/models/gguf/llama.gguf \
  --mmproj /opt/models/gguf/mmproj-Qwen_Qwen3.5-35B-A3B-f16.gguf \
  --host 0.0.0.0 --port 11434 \
  --ctx-size 262144 -np 1 -fa on \
  -b 1024 -ub 512 --threads 16 \
  --cache-type-k q4_0 --cache-type-v q4_0 \
  --jinja --reasoning off --reasoning-budget 0 \
  --metrics
```

Revision History

Version	Date	Changes
1.0	2026-04-05 23:32 UTC	Initial document
1.1	2026-04-05 23:56 UTC	Added metrics pipeline docs, cron automation, llama-server config, revision history

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